DBA5103: Project Guideline
Semester I, 2026/2027, NUS

1 Overview of Term Project

Each project team is required to prepare a term paper on a practical problem or a research topic relevant to the subjects covered in this class. Any paper used for fulfilling requirements of other courses must not be recycled in this class. It is allowed, however, to have the same problem background or data set but with a different focus.

The purpose of the term paper is to demonstrate that you can apply the analytical techniques learned in this class to a practical problem or a research topic of your choosing. The paper must include a motivation and statement of the problem, data or process(es) analyzed, a clear definition of the mathematical model, and the methodology used or principles learned. Codes used to implement the problem should be attached as appendix.

The paper should be typewritten, paginated, double-spaced, in Time font, size 12, 1 inch margins (top, bottom, left, and right). Other formats, as long as consistently applied and cleanly convey the content, are also acceptable. There is no strict page limitation, but a good term paper (exluding the appendix) may need 8 to 14 pages of narratives to provide in-depth analysis of a selected topic. Note, however, that a longer term paper does NOT mean better. A concisely written term paper with one sharp point in, say, 6 pages is far better than a 30 pages of report with no single clear point and filled with paragraphs copied and pasted from various other sources.

2 Topics for Term Project

As mentioned in the overview, generally speaking, there are two possible types of project. One is practical problem oriented. The other one is research topic oriented. The different focuses are elaborated below with examples given.

- **Practical Problem Oriented**

  For this type, the problem can either come from reality or virtual reality. The central question in this type of project is simply to provide a solution to a particular problem. The key issue, however, is to connect the model that is built to derive the solution and that of the practice. For example, what are the assumptions made in the model that can or cannot capture the reality; are the solutions robust to a change of environment or assumptions; can a better model and solution be proposed if more information is available.

  One prototype example could be the markdown management problem discussed in class. By collecting past sales data, one can form a point estimate of the demand at each price point. Through imposing the assumptions that (1) demand can be perfectly forecasted; (2) inventory decisions are given in advance; (3) only four price points are available; one can formulate a simple LP to solve for the optimal pricing decisions. One then reflect on the assumptions made: (1) if demand happens to be not the same as the forecast, then one can use a rolling-horizon strategy to dynamically adjust the decisions; (2) one can probably resolve the model for different inventory levels and then determine the best amount of stocks to keep before the selling season; (3) one can resolve the model with more price points and examine the possible impact on the profit. With all elements combined together, this could be a good project.

Some good examples from previous years are given below.

Determining the porter station locations in a hospital. The legacy system is 11 years old with new medical departments being built up and the current porter station system is unable to meet the demand from medical departments. Hence, new locations for porter stations are desirable. The team collected data, estimated demands for porter service from different medical departments, traveling time between two locations and capacity level of the porter stations. A facility location + transportation type of model is built to model the problem. The problem is recognized to be an Mixed Integer Programming (MIP) problem. Since demands in practice could be random, and capacity as well as the number of the stations can be changed at additional cost, extensive sensitivity analysis with respect to these assumptions are conducted, and clear policy implications are summarized.

Integer programming approach in solving Flow-Free game puzzle. Flow-free game is a combinatorial game that can be played on mobile phones (see `https://itunes.apple.com/sg/app/flow-free/id526641427?mt=8`). The player needs to connect matching colors with pipe to create a Flow. To complete the puzzle, one needs to pair all colors, and cover the entire board without letting any pipes to cross or overlap. The team developed an integer programming (IP) model to solve the puzzle including the state of the art 14×14 large size puzzle that is available in the game consoles. The IP model is further extended to address the more complicated Flow-Free games with various additional constraints. The complexity of the resulting model, e.g., number of constraints, number of variables, are discussed in detail and the computational time is also reported in the numerical studies.

Manufacturing of Complex Machine. The team models the manufacturing process of a complex machine as a project management problem, in which the activities/steps in the project is interpreted as the production/procurment of a component for the machine. Real data is collected to provide the information on the production/procurement lead time for each of the components and the corresponding expediting cost/outsourcing cost. The project management model used in class is generalized to include the binary decision of whether to outsource the production of a component to an external supplier or not. Further practical elements such as random lead time, multiple suppliers to outsource from are also considered in the extension.

Other topics from previous years that fall into this category include vessel route optimization with fuel price consideration (TSP type of problem); scheduling module time and classroom for the students (scheduling type of problem); locating childcare centres in HDB blocks in Singapore (facility location type of problem); maximizing fantasy points in fantasy football game (resource allocation type of problem).

- **Research Topic Oriented**

  The central question in this type of project is more concerned with a general methodology rather than a particular practical instance. Real data is not necessarily required to answer this type of question but the question itself is usually more sophisticated compared to the central question in the first type of project. Reference is far more important in this type of project (it is important in the first type too). Is the question you are asking studied before? What are the previous methods in answering the questions? Novel and new discovery is, of course, encouraged but not necessary. A summary and reproduction of what are already discovered can also be nice, but, again, proper reference should be given.

  One prototype example could be the process flexibility problem discussed in class. The central question there is: what is a good design of flexible production when facing random demands? For a fixed demand realization, the problem can be formulated as a network flow problem and combined with simulation one can evaluate the performance of a design given a demand distribution. Two benchmarks: dedicated system and full flexible system are evaluated. In the literature, it is proposed that open chain and long chain are possible designs.

Using the optimization model, one can evaluate the two designs and consistent findings are obtained. One can then further answer relevant questions: Do the findings still hold in unbalanced system or for other demand distributions? What is the relationship between the performance of long chain and the variance in demand? How can we design optimal flexibility structure if adding an arc in the graph comes with certain cost?

Some good examples from previous years are given below.

Stable matching problem from a linear programming perspective. The central question of stable matching problem (or stable marriage problem) concerns with how to find a matching of people/entities (say, one man to one woman or one medical school graduate to one hospital) such that no two couples will feel happier by exchanging their partners. The well-known algorithm for this problem is called Gale-Shapley algorithm. The team examines the approach of using linear programming to solve the problem. They discussed in detail the findings from the literature that how to reformulate the problem as LP, what the key constraint is in ensuring a stable matching, and confirmed the result that by solving a transportation type of problem one can obtain a stable matching. They further examine how a different objective function can result in a different stable matching.

Optimal Classification Tree. The CART (classification and regression tree) method in traditional statistical learning is based on a greedy heuristic that works top-down by choosing a variable at each step that best splits the set of items. The resulting tree from CART, however, is usually not the best possible tree. The team then studied and reproduced the integer programming formulation proposed in Bertsimas and Dunn (2017): Optimal Classification Trees, *Machine Learning*, 106(7), 1039–1082. The team further tested the performance of the tree obtained from solving the integer programming model and compared that with CART for several data sets.

Assignment optimization under uncertain demand. The project was motivated from the Uber's dispatch system discussed in class. In particular, it studies the tradeoff between the greedy approach where a customer is immediately assigned to the nearest driver whenever he/she arrives and the perfect information approach where the platform waits until all demands arrive and then finds the optimal assignment between drivers and customers. Since demands are random, one will not have the information of all demands in advance, and hence the perfect information approach is not practical. The team examines the intermediate case where the platform waits for some periods and match realized demands with supply in batches. A simulation model based on synthetic data is built to evaluate the performance of the algorithm. By considering different waiting periods before an assignment is made, a trade-off between the greedy approach and the perfect information approach is demonstrated.

Other topics from previous years that fall into this category include comparison of L_2, L_1 and Chevyshev criterion in regression problems; comparison of four heuristics in a large scale TSP problem; finding a way to improve upon the K-means algorithm in a clustering problem.

3 Evaluation of Term Project

The evaluation is mainly based on the following five aspects:

- **Understanding of the Topic**

  This includes: whether the problem at a hand is modeled properly; whether the model is solved using a proper algorithm; whether the model/algorithm is related to some more general model/algorithm studied in the literature for similar problems; whether the corresponding references are cited properly.

- **Overall Writing**

This includes: whether the flow is well-structured; whether the main points of the project clearly conveyed; whether the notations are used in a consistent and professional way; whether the data, figures, tables are explained clearly.

- **Definition of the Problem**

  This includes: whether the optimization problem is clearly defined; whether each constraint is properly explained; whether the type of optimization problem is recognized.

- **Methodology**

  This includes: whether the algorithm or method is properly explained (especially if it is not using a standard method discussed in class); whether the algorithm or method is sound and correct.

- **Result Analysis**

  This includes: whether outputs from the algorithms are explained; whether the implications/recommendations/limitations are discussed.